

Chenyang (Danny) Ma

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EDUCATION

University of Oxford <i>Ph.D. in Computer Science. Focus: Embodied AI, Robotics, Agents</i>	Oct 2023 – July 2027 (Expected)
University of Cambridge <i>MPhil in Advanced Computer Science</i>	Oct 2022 – July 2023 Distinction
University of Michigan <i>B.S.E. in Computer Engineering, Minor in Mathematics</i>	Sept 2019 – May 2022 Cumulative GPA: 3.97

RESEARCH & INDUSTRY EXPERIENCE

PhD Student at CPS Group, University of Oxford <i>Mentors: Andrew Markham & Niki Trigoni</i>	Oct 2023 – Present Oxford, UK
- SpatialPIN — a modular plug-and-play framework that progressively enhances VLM’s 3D reasoning capabilities by prompting and interacting with 3D foundational models - COOPERA — a framework for continual and open-ended human-robot collaboration - CycleVLA — a proactive self-correcting VLA that predicts and recovers from failures	
Research Scientist Intern at Mitsubishi Electric Research Laboratories <i>Mentors: Diego Romeres & Chiori Hori</i>	Apr 2025 – Apr 2026 Cambridge, US
- Introduced FurnitureVLA — learning a generalist robot policy for real-scale, long-horizon bimanual furniture assembly with VLA - Co-developed KitchenVLM — a zero-shot error-correction framework for generating and optimizing executable robot actions from human instructional videos	
Research Scientist Intern at IBM Research <i>Mentor: Dave Braines</i>	June 2025 – Sept 2025 Hursley, UK
- Developed EmbeWebAgent — a lightweight and stack-agnostic framework for embedding web agents into enterprise legacy UIs using minimal frontend hooks and a reusable backend workflow - Enabled mixed-granularity actions, explicit nested navigation, and multi-agent orchestration with session-scoped memory for robust multi-step action execution in real-world enterprise environments - Integrated the framework into IBM’s Safer Materials Advisor project; to be released in IBM MVP 4.0	
Research Collaborator at FAIR, Meta <i>Mentors: Xavier Puig & Ruta Desai</i>	June 2024 – Apr 2025 Remote / SF Bay Area, US
- Proposed COOPERA, a framework enabling the first study of continual, open-ended human-robot collaboration - Developed a method to simulate realistic humans within robot simulation software using LLMs and motion data - Introduced a benchmark and an approach to personalize robot actions through multiple days of collaboration	
Applied Scientist Intern at Roku <i>Mentor: Michael Sanders</i>	July 2024 – Oct 2024 Cambridge, UK
- Investigated the problem of IoT camera package delivery detection under adversarial conditions - Framed the problem within and addressed pain points in video-based human action recognition caused by the complexity of real-world data (e.g., high variety of camera angles, backgrounds, illumination, scales, etc.) - Developed a cost-effective hierarchical ensemble pipeline with a meta-learner to amplify human action signals and learn causal relationships between models, achieving performance ready for real-world deployment	
Research Scientist Intern at Flower Labs & CaMLSys, University of Cambridge <i>Mentors: Nicholas Lane & Daniel Beutel</i>	Oct 2022 – July 2023 Cambridge, UK
Developed the first privacy-preserving framework for federated XGBoost under horizontal federated learning setting that does not depend on the sharing of gradients and Hessians, which leads to serious privacy concerns	

- Proposed a novel method to transform the tree ensembles built by local clients as inputs to neural networks to learn robust learning rate strategies

Research Assistant at Owens Lab, University of Michigan

July 2021 – Sept 2022

Mentors: Andrew Owens & Wenzhen Yuan

Ann Arbor, US

- Established Touch and Go — a human-collected visual-tactile dataset with 4000 different real-world objects, 14 hours of videos, and 13,900 touches which enables researchers to study diverse visual-tactile learning applications beyond the robotics-centric domains
- Applied our dataset on multimodal learning tasks including self-supervised tactile-visual representation learning, tactile-driven image stylization, and multimodal future touch prediction

Student Researcher at University of Michigan

Jan 2021 – Feb 2022

- Conducted research on semantic segmentation for remote sensing images by alleviating large intra-class variance in both foreground and background classes
- Constructed a sparse and complete latent structure via prototypes to tackle the above issues by designing a prototypical contrastive learning strategy and modeling all foreground and hardest background objects
- Designed a novel patch shuffle augmentation to encourage the semantic information of an object to be correlated only to the limited context within the patch that is specific to its category

SELECTED PUBLICATIONS

FurnitureVLA: Learning Long-Horizon Bimanual Furniture Assembly with Vision-Language-Action Model. Preprint, 2026

Chenyang Ma, Yue Yang, Radu Corcodel, Siddarth Jain, Andrew Wu, Chiori Hori, Diego Romeres

CycleVLA: Proactive Self-Correcting Vision-Language-Action Models via Subtask Backtracking and Minimum Bayes Risk Decoding. Preprint, 2026

Chenyang Ma, Guangyu Yang, Kai Lu, Shitong Xu, Bill Byrne, Niki Trigoni, Andrew Markham

EmbeWebAgent: Embedding Web Agents into Any Customized UI. Technical Report, 2026

Chenyang Ma, Clyde Fare, Matthew Wilson, Dave Braines

COOPERA: Continual Open-Ended Human-Robot Assistance. NeurIPS, 2025. [Selected as Spotlight](#)

Chenyang Ma, Kai Lu, Ruta Desai*, Xavier Puig*, Andrew Markham*, Niki Trigoni* (*= Equal Advising)

SpatialPIN: Enhancing Spatial Reasoning Capabilities of Vision-Language Models through Prompting and Interacting 3D Priors. NeurIPS, 2024

Chenyang Ma, Kai Lu, Ta-Ying Cheng, Niki Trigoni, Andrew Markham

Gradient-less Federated Gradient Boosting Tree with Learnable Learning Rates. EuroMLSys Workshop, 2023

Chenyang Ma, Xinchu Qiu, Daniel Beutel, Nicholas Lane

Touch and Go: Learning from Human-Collected Vision and Touch. NeurIPS, 2022

Fengyu Yang*, Chenyang Ma*, Jiacheng Zhang, Jing Zhu, Wenzhen Yuan, Andrew Owens (*= Equal Contribution)

Sparse and Complete Latent Organization for Geospatial Semantic Segmentation. CVPR, 2022

Fengyu Yang*, Chenyang Ma* (*= Equal Contribution)

ACADEMIC SERVICES

- Reviewer:** ECCV (2026-), ICCV (2025-), CVPR (2023-), NeurIPS (2022-)
- Teaching Assistant:** Oxford Computer Arch. (2026), DL in Healthcare (2024), Machine Learning (2023)

HONORS & AWARDS

- Best Teaching Assistant Nomination**, Department of Computer Science, University of Oxford 2024
- Summa Cum Laude**, College of Engineering, University of Michigan 2022
- James B. Angell Scholar**, College of Engineering, University of Michigan 2021
- Dean's List**, College of Engineering, University of Michigan 2019 - 2022
- University Honors**, University of Michigan 2019 - 2022